Correction: Uniform Vorticity Envelope versus Cubic Differential Hypothesis

Frohman, Benjamin Stanley
frohmanbenjamin@gmail.com

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Abstract

This addendum separates the uniform-envelope assumption in the checked Lean reduction from a genuinely differential assumption on the actual vorticity supremum. It records the conditional comparison proved in the accompanying source patch, tests a proposed coercivity mechanism on explicit periodic solutions, and identifies the remaining analytic obligation. It is not a proof of general Navier–Stokes regularity. Earlier published PDFs and archives are unchanged.

1 The corrected distinction

The baseline source is `BenFrohman/NS_Millennium_Proof` at commit `c98fe859867affeba4508b3e51583cfa6b20087c`. Historically `DirectionCurvatureBound` abbreviates `PeriodicDirectionDepletionBound`. Its actual body assumes that, for every smooth divergence-free periodic datum, there are positive constants C, κ such that every represented local solution satisfies

$$M(t) \le \max\{M(0), C/\kappa\}, \qquad 0 \le t < T.$$

Here $M(t) = \|\operatorname{curl} u(t)\|_\infty$. The constants may depend on viscosity and the datum, but not on the solution or its lifespan. This definition does not assert a differential inequality. The accurate public name is now `UniformVorticityEnvelopeBound`; the original body, historical aliases and original conditional theorem are preserved.

The separate `CubicVorticityDifferentialBound`(u, τ, C, κ) explicitly requires $\tau \ge 0$, $C, \kappa > 0$, bounded spatial vorticity-norm ranges, ordinary differentiability of the actual M at every point of $[0, \tau]$, and

$$M'(t) \le CM(t)^2 - \kappa M(t)^3 \quad (0 \le t \le \tau).$$

The boundedness guard matters for the real-valued conditional supremum used in Lean. No differential estimate or differentiability of a supremum is inferred merely from smoothness of the velocity.

2 The checked conditional consequences

The existing `comparison_ode_ceiling`, with all its hypotheses, gives the compact envelope above. The new source proves this for the actual curl supremum, then bounds the actual curl pointwise using spatial boundedness. On an actual `IsPeriodicSmoothSolutionOn`, compatible compact hypotheses for every $0 \le \tau < T$ with the *same* coefficients give the envelope on $[0, T)$; the

initial identity comes from that solution's definition. A zero-trajectory regression shows that the differential predicate is inhabited.

`clay_reduces_to_uniform_vorticity_envelope` accurately names the existing conditional reduction: the uniform envelope for every positive viscosity, *and* the separately assumed periodic BKM continuation principle, imply `Clay_B_exact`. Neither premise is established by that reduction.

The Stage6 additions check the derivative of $M_0 e^{-at}$, its cubic estimate with $C = \kappa M_0$, and the fixed-coefficient obstruction below. These are *scalar* calculus and algebra theorems. Assembly of the following vector fields into the Lean solution structure and identification of their suprema are not newly formalized here; the explicit PDE calculations are separate mathematical checks.

3 What the PDE identity supplies

For a classical unforced incompressible solution, let $\omega = \operatorname{curl} u$, $S = (\nabla u + (\nabla u)^T)/2$ and $\rho = |\omega|$. Then

$$(\partial_t + u \cdot \nabla)\frac{|\omega|^2}{2} = \omega \cdot S\omega + \nu\Delta\frac{|\omega|^2}{2} - \nu|\nabla\omega|^2.$$

Where $\rho > 0$, write $\omega = \rho\xi$, $|\xi| = 1$. Differentiating gives

$$|\nabla\omega|^2 = |\nabla\rho|^2 + \rho^2|\nabla\xi|^2,$$
$$(\partial_t + u \cdot \nabla - \nu\Delta)\rho = \rho\,\xi \cdot S\xi - \nu\rho|\nabla\xi|^2.$$

At a positive spatial maximum $\nabla\rho = 0$ and $\Delta\rho \le 0$. These facts do not give a lower bound on direction curvature. A general maximum-in-time argument also needs a justified one-sided/Dini derivative treatment of maximizing-branch switches, not an automatic ordinary derivative of M.

3.1 Beltrami family: fixed coefficients versus data dependence

On the unit torus take $\nu > 0$, $t \ge 0$, $k = 2\pi$, $a = \nu k^2$, $A > 0$, and

$$u_A(t,x) = Ae^{-at}(\sin(kx_3), \cos(kx_3), 0), \qquad p = 0.$$

Direct differentiation gives $\operatorname{div} u_A = 0$, $(u_A \cdot \nabla)u_A = 0$, $\Delta u_A = -k^2 u_A$, $\partial_t u_A = \nu\Delta u_A$ and $\operatorname{curl} u_A = k u_A$. Consequently

$$M_A(t) = kAe^{-at}, \qquad M_A' = -aM_A.$$

At zero, the cubic estimate is equivalent to $\kappa M_A(0)^2 - CM_A(0) - a \le 0$. For fixed $C \ge 0$, $\kappa > 0$ and $a \ge 0$, this fails at sufficiently large amplitude. The Lean theorem `fixed_cubic_constants_fail_for_some_amplitude` proves the quantified scalar obstruction. This does *not* refute the data-dependent envelope: choosing $C = \kappa M_A(0)$ makes the cubic right-hand side $\kappa M_A(t)^2(M_A(0) - M_A(t)) \ge 0$, while $M_A' \le 0$. The latter special-case sufficient estimate is also checked in Lean for the scalar profile.

3.2 Single shear: direction-only coercivity fails

Instead take the actual periodic heat shear

$$v_A(t,x) = Ae^{-at}(\sin(kx_3), 0, 0), \qquad p = 0.$$

The same divergence, convection and heat calculation verifies the unforced Navier–Stokes equations. Here

$$\omega_A(t,x) = (0, kAe^{-at}\cos(kx_3), 0).$$

At $x_3 = 0$, its norm attains the positive global maximum $M = kAe^{-at}$. In a neighborhood $\cos(kx_3) > 0$, so $\xi = (0,1,0)$ is constant and $|\nabla\xi|^2 = 0$. Thus the bare premise

$$\kappa M^2 \le \nu|\nabla\xi|^2$$

from `viscous_direction_dissipation_bridge` fails for *every* $\kappa > 0$, even if it depends on this datum. This refutes promotion of that premise to an unconditional assertion at every positive maximum, not the existing implication which assumes it. Nor does it refute a different estimate restricted above a data-dependent threshold. The actual damping here comes from amplitude curvature:

$$\omega_A \cdot \Delta\omega_A = -k^2M^2, \qquad M' = -\nu k^2 M.$$

Discarding amplitude curvature loses this real viscous contribution.

3.3 Scaling is not a coercivity proof

For the parabolic rescaling $u_\lambda(t,x) = \lambda u(\lambda^2 t, \lambda x)$, one has $M_\lambda(t) = \lambda^2 M(\lambda^2 t)$ and the time derivative scales by λ^4 wherever it exists. A cubic inequality therefore transforms with $C_\lambda = C$ and $\kappa_\lambda = \kappa/\lambda^2$, not fixed positive κ. Likewise $\nu|\nabla\xi|^2$ scales by λ^2 whereas M^2 scales by λ^4. This does not contradict data-dependent coefficients. On the fixed unit torus the displayed transformation preserves unit periodicity for positive integer λ; arbitrary domain rescalings are not the same torus. The separate scale-homogeneous premise $\kappa_{\text{eff}} M \le \nu|\nabla\xi|^2$ in the source only yields $M' \le (C - \kappa_{\text{eff}})M^2$, not cubic damping in M. Scaling supplies neither coercivity premise.

4 Endpoint representation and remaining obligation

The Lean local-solution structure constrains only nonnegative times. Extend the zero forward solution by $(\sin(kx_3), \cos(kx_3), 0)$ for negative times. It is still the zero solution on $[0,T)$, but its actual vorticity supremum jumps from k to zero at time zero. Hence ordinary two-sided differentiability of M at zero cannot follow for every represented solution. This is not an obstruction to right-sided PDE smoothness. The globally zero trajectory, unlike this other extension, passes the new regression. The compact differential predicate is an explicit additional condition on a given trajectory, not a universal PDE theorem.

A sufficient net-source estimate at the relevant positive maxima would be

$$\omega \cdot S\omega + \nu\omega \cdot \Delta\omega \le CM^3 - \kappa M^4,$$

with correctly quantified positive coefficients uniform in time and lifespan. Division by M then gives the cubic rate after a justified maximum-derivative argument. The repository proves the algebra conditionally, not this general source estimate. An endpoint L^∞ Calderon–Zygmund bound cannot silently supply it. A suitable source estimate and a one-sided/Dini comparison bridge remain analytic/formal obligations. No general regularity breakthrough is claimed.

5 Reproduction and evidence boundary

Use Lean `leanprover/lean4:v4.30.0-rc1` and Mathlib commit `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7`; all package pins are in `lake-manifest.json`. Run `lakebuild`, then the Lean scripts `scripts/AxiomAudit.lean`, `scripts/EnvelopeCorrectionAudit.lean`, and `scripts/CertificateAxiomInventory.lean` using `lakeenvlean`. The correction audit checks old API compatibility and every new theorem's axioms; the inventory reports transitive axioms per noninternal project theorem. No new axiom, `sorry` or `native_decide` is introduced. Standard-axiom checks establish neither a missing analytic hypothesis nor novelty.

The Markdown companion `docs/VORTICITY_ENVELOPE_CORRECTION.md` supplies exact pinned source permalinks and the full named-theorem list. The review bundle contains sources, dependency/toolchain pins, logs, project artifacts and hashes; it is not a standalone independently trusted proof certificate. Earlier published PDFs/Zenodo records are not modified or republished.

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